CBRNet: Corner-Guided Boundary Refinement Network for High-Precision Building Extraction From Remote Sensing Imagery

Min Huang, Shuaishuai Cao, Daoye Zhu, Xuan Liu, Jin Luo, Yong Chen, Jiachen Niu, Li Zhang, Xiaoyi Huang, and Hui Lin

***Abstract*—Accurate building extraction from high-resolution remote sensing imagery remains essential for urban planning, disaster assessment, and geographic information system updates. Existing methodologies face three critical challenges: 1) substantial scale variations across diverse urban contexts; 2) boundary ambiguity induced by shadows and spectral similarities; and 3) progressive loss of fine-grained structural details during hierarchical feature aggregation. We propose CBRNet, a corner-guided boundary refinement network that introduces structural geometric reasoning into deep building extraction. Our key insight is that building corners serve as anchor points for boundary delineation—by detecting corners first, boundary precision naturally follows. CBRNet features four synergistic innovations: 1) content-adaptive scale convolution (CASC) that dynamically adjusts receptive fields based on local content; 2) corner-guided feature enhancement (CGFE) that integrates differentiable Harris corner detection as learnable geometric priors; 3) local window cross-attention (LWCA) enabling cross-scale interaction with linear complexity; and 4) dual-stream boundary refinement (DSBR) that decouples semantic and boundary learning with bidirectional feature exchange. In addition, we employ bidirectional feature pyramid network (BiFPN) for efficient multiscale feature aggregation. Extensive experiments demonstrate state-of-the-art results: 91.35% IoU on WHU Aerial, 76.86% on HUST-ABS, and 84.28% on WHU Satellite I at 512 × 512 resolution. Notably, CBRNet generalizes robustly to 1024 × 1024 resolution on Inria, achieving 81.87% IoU without architectural modification. CBRNet achieves 79.8 FPS with only 32.81-M parameters, establishing an optimal accuracy-efficiency tradeoff for real-world deployment.**

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I. Introduction

BUILDING extraction from high-resolution remote sensing imagery constitutes a fundamental task within the geospatial intelligence community, enabling extensive applications including urban planning [1], disaster damage assessment [2], population estimation [3], and topographic map updating [4]. The proliferation of high-resolution satellite and aerial imaging platforms has provided unprecedented opportunities for detailed urban monitoring; however, it has simultaneously introduced substantial technical challenges that necessitate sophisticated computational approaches [5]. The inherent complexity of urban environments, characterized by heterogeneous building morphologies, varying imaging conditions, and diverse geographic contexts, poses significant obstacles for automated building extraction systems [6].

The past decade has witnessed remarkable progress in building extraction methodologies, transitioning from traditional image processing techniques to deep learning-based semantic segmentation frameworks [7], [8]. Convolutional neural networks (CNNs), particularly encoder–decoder architectures such as U-Net [9] and its variants including U-Net++ [10], U-Net 3 + [11], and Attention U-Net [12], have established dominant paradigms by effectively capturing hierarchical spatial features through progressive downsampling and upsampling operations. More recently, transformer-based models [8], [13] and foundation models such as the segment anything model (SAM) [14] and its successors SAM 2 [15] and SAM 3 [16] have demonstrated superior capability in modeling long-range dependencies and zero-shot generalization, further advancing state-of-the-art performance across various benchmarks.

Despite these remarkable achievements, three fundamental challenges remain inadequately addressed in existing approaches, significantly limiting their practical deployment in real-world scenarios as follows.

Challenge 1: Scale Variation: Buildings exhibit substantial scale variations across different urban environments, ranging from small residential structures occupying merely tens of pixels to large industrial complexes spanning thousands of pixels [17]. Fixed-scale convolutional kernels inherently struggle

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to accommodate such diversity, frequently resulting in fragmented predictions for large buildings due to limited receptive fields or erroneous merging of adjacent small structures due to over-smoothing effects. Although multiscale feature extraction strategies such as atrous spatial pyramid pooling (ASPP) [7] and feature pyramid networks (FPNs) [18] partially mitigate this limitation, they typically employ static, content-agnostic scale selection mechanisms that fail to adapt to the varying characteristics of different image regions.

Challenge 2: Boundary Ambiguity: Building boundaries frequently suffer from severe ambiguity due to shadows cast by adjacent structures, vegetation occlusion, and spectral similarity with surrounding objects such as roads and parking lots [19]. Conventional segmentation networks optimize for overall pixel accuracy, which inadvertently biases the learning process toward interior regions that constitute the majority of building pixels while neglecting boundary precision. This phenomenon is particularly problematic for downstream applications such as vectorization, cartographic generalization, and area calculation, where boundary accuracy directly determines the utility of extraction results [20].

Challenge 3: Structural Detail Loss: The progressive downsampling operations inherent in encoder networks inevitably cause the loss of fine-grained structural details, including building corners, edges, and geometric regularities that characterize man-made structures [21]. While skip connections in U-Net-style architectures attempt to preserve spatial information, they provide limited capability to explicitly model geometric priors that distinguish buildings from natural objects. Consequently, the reconstructed building footprints often exhibit irregular boundaries, rounded corners, and distorted shapes that deviate from the inherent geometric properties of architectural structures.

To comprehensively address these interrelated challenges, we propose corner-guided boundary refinement network (CBRNet), a novel deep learning architecture that synergistically integrates corner-guided structural enhancement with dual-stream boundary refinement (DSBR) within a unified end-to-end trainable framework. The core design philosophy of CBRNet centers on two fundamental geometric priors inherent in building structures: **corner prominence** (buildings are characterized by distinctive corner vertices that serve as structural landmarks) and **boundary regularity** (building edges exhibit geometric consistency that distinguishes them from natural objects). By explicitly modeling these priors through dedicated network modules, CBRNet achieves superior extraction accuracy while maintaining adaptability to diverse imaging conditions.

The principal contributions of this article are summarized as follows.

1) We propose the content-adaptive scale convolution (CASC) module, which dynamically predicts optimal receptive field scales for each spatial location based on local feature content. By synergistically combining global context prediction with local refinement, CASC enables adaptive multiscale feature extraction that automatically adjusts to varying building scales within individual images.
2) We introduce the corner-guided feature enhancement (CGFE) module, which explicitly incorporates differentiable Harris corner detection into the deep learning pipeline as a structural prior. By computing corner response maps from learned feature representations and utilizing them to guide attention learning, CGFE emphasizes geometrically salient regions corresponding to building corners and edges.
3) We develop the local window cross-attention (LWCA) mechanism, which captures fine-grained spatial dependencies through cross-scale feature interaction within local windows while maintaining linear computational complexity with respect to image size.
4) We design the DSBR decoder, which separately models semantic regions and boundary details through dedicated processing streams with bidirectional cross-stream interaction for mutual enhancement.
5) We conduct extensive experiments on four challenging building extraction benchmarks with varying spatial scales (from 512×512 to 1024×1024), demonstrating that CBRNet achieves state-of-the-art performance while maintaining competitive computational efficiency.

The remainder of this article is organized as follows. Section II reviews related work on building extraction and relevant deep learning techniques. Section III presents the detailed architecture of CBRNet and its constituent modules. Section IV describes experimental settings and comprehensively analyzes the results. Section V concludes this article with discussions on limitations and future research directions.

II. Related Work

A. Deep Learning for Building Extraction

Deep learning-based building extraction has evolved through several paradigm shifts over the past decade. Early approaches adopted fully convolutional networks (FCNs) [26] to enable end-to-end pixel-wise classification. The introduction of encoder–decoder architectures, exemplified by U-Net [9], established the foundational framework for dense prediction tasks. Subsequent variants, including U-Net++ [10] with nested skip connections, U-Net 3 + [11] with full-scale feature aggregation, and Attention U-Net [12] with attention gating mechanisms, have further enhanced performance through improved feature propagation.

Recent advances have focused on multiscale representation learning and attention mechanisms. PSPNet [27] proposed pyramid pooling modules for global context aggregation, while the DeepLab series [7] introduced atrous convolutions to expand receptive fields. HRNet [28] maintained high-resolution representations throughout the network. For building extraction specifically, MCCANet [22] designed multiscale cross-channel attention modules, CMTFNet [23] explored cross-modal transformer fusion, and BuildFormer [24] adapted vision transformers (ViT) for building extraction. DBBANet [19] introduced dual-branch boundary-aware attention mechanisms, while Mask2Former [25] unified various segmentation tasks through masked attention. More recently, He et al. [54] proposed UANet, an uncertainty-aware network

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that explicitly models prediction uncertainty to improve building extraction robustness, and Li et al. [53] further addressed uncertainty challenges in the context of building change detection from multitemporal remote sensing imagery.

The emergence of foundation models has introduced new paradigms for image segmentation. The SAM [14] demonstrated remarkable zero-shot generalization through large-scale pretraining. SAM 2 [15] extended this capability to video segmentation with memory-based propagation, while SAM 3 [16] introduced concept-aware segmentation for enhanced semantic understanding. However, these general-purpose models often require task-specific adaptation for optimal building extraction performance.

B. Multiscale Feature Learning

Multiscale feature learning addresses the fundamental challenge of scale variation in visual recognition. The FPN [18] established a top-down pathway for propagating semantic information. PANet [30] augmented this with an additional bottom-up path. BiFPN [29] introduced learnable weighted fusion for efficient multiscale combination.

Recent innovations have explored adaptive mechanisms for scale selection. Deformable convolutions [31], [32] learn spatial offsets that adjust receptive field shapes according to object geometry. SKNet [33] introduced kernel selection attention for adaptive kernel size selection. Dynamic convolution [34] conditions the entire feature extraction process on input content. Unlike these approaches, our proposed CASC module combines global context prediction with pixel-wise local refinement for fine-grained scale adaptation.

C. Boundary-Aware Segmentation

Boundary quality has emerged as a critical consideration for segmentation accuracy. Early boundary-aware methods incorporated edge detection as auxiliary supervision [35], while subsequent approaches developed specialized boundary prediction branches [36]. SegFix [37] proposed postprocessing refinement that propagates predictions from interior regions to boundaries.

For building extraction, boundary awareness assumes heightened importance due to the geometric regularity inherent in man-made structures. BRRNet [38] introduced boundary-guided refinement through explicit boundary supervision. MAP-Net [39] formulated multiscale attention with boundary supervision. Recent works, including PolyBuilding [40] and PolyWorld [20], have explored polygonal building extraction for enhanced boundary precision. Our DSBR module advances this direction by establishing bidirectional information flow between semantic and boundary streams.

D. Foundation Models and Large-Scale Pretraining

The success of large language models has inspired the development of foundation models for computer vision. The ViT [41] demonstrated the effectiveness of transformer architectures for image recognition. Swin transformer [42] introduced local window attention with shifted windows for efficient hierarchical representation learning.

The SAM [14] represents a significant advancement in promptable segmentation, trained on over 1 billion masks from 11 million images. SAM 2 [15] extended this capability with streaming memory for video understanding. Most recently, SAM 3 [16] introduced concept-level understanding through a multitask framework that integrates object detection, segmentation, and captioning. While these foundation models provide strong general-purpose capabilities, domain-specific architectures like our CBRNet can leverage geometric priors inherent in building structures for enhanced extraction accuracy.

Beyond building extraction, self-supervised learning, and low-rank representation techniques have demonstrated remarkable success in addressing various remote sensing image processing challenges. Chen et al. [50] proposed a coupled low-rank subspace representation with self-supervised deep networks for hyperspectral compressive snapshot reconstruction, achieving state-of-the-art reconstruction quality. The same research group extended these principles to thick cloud removal in multitemporal remote sensing images through low-rank regularized self-supervised networks [51], effectively recovering obscured ground information. More recently, Yuan et al. [52] developed feature fusion-guided networks with sparse prior constraints for unsupervised hyperspectral image quality improvement. These advances in self-supervised and unsupervised learning paradigms provide valuable insights for developing more robust building extraction methods under challenging imaging conditions.

III. Methodology

A. Overall Architecture

The overall architecture of CBRNet is illustrated in Fig. 1. CBRNet adopts a sophisticated encoder–decoder structure with ResNet-50 [43] serving as the backbone encoder, extracting hierarchical features at four scales with stride factors of 4, 8, 16, and 32 relative to the input resolution.

Specifically, for deep-level features C_3 and C_4 that encapsulate rich semantic information essential for distinguishing buildings from background, we sequentially apply: 1) CASC for dynamic multiscale feature extraction; 2) CGFE for structural emphasis that leverages geometric priors; and 3) LWCA for fine-grained spatial reasoning. The enhanced multiscale features are subsequently aggregated through a bidirectional FPN (BiFPN) [29] to produce unified representations at each scale. The highest-resolution feature map from BiFPN is fed into the DSBR decoder, which generates the final semantic segmentation and boundary predictions.

B. Content-Adaptive Scale Convolution

Buildings in remote sensing imagery exhibit substantial scale variations that fundamentally challenge conventional fixed-scale convolutional operations. To address this limitation, we propose the CASC module that dynamically determines optimal receptive field scales based on local feature content.

Given an input feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ from the backbone encoder, CASC first applies K parallel depthwise-separable

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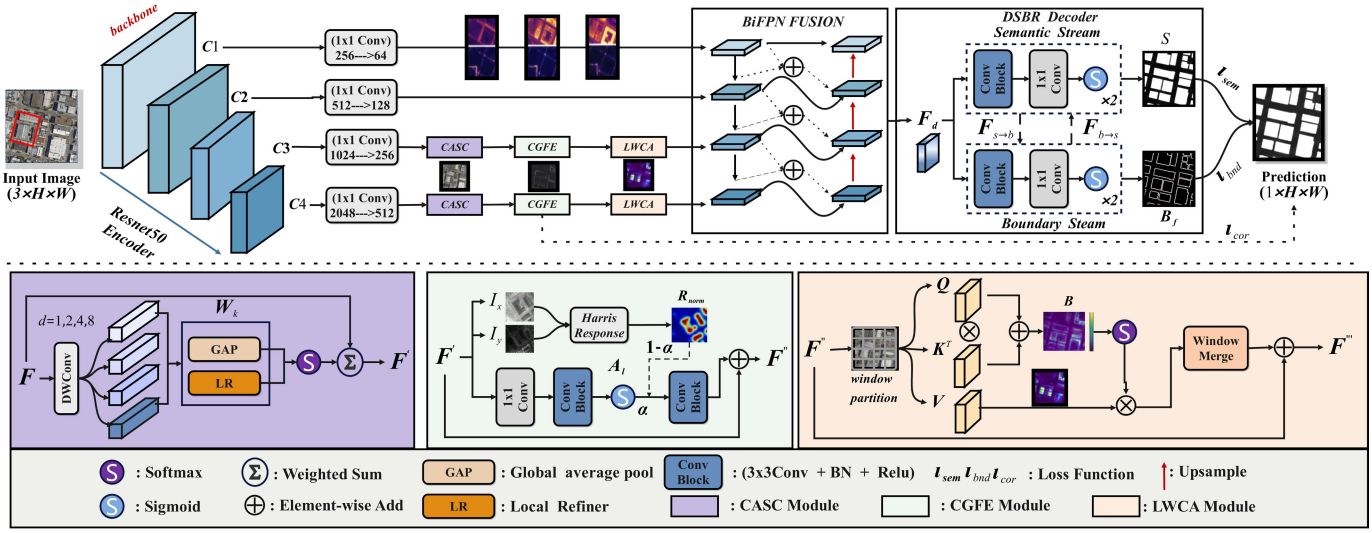


Fig. 1. Overall architecture of the proposed CBRNet. The encoder extracts hierarchical features at multiple scales with stride factors of 4, 8, 16, and 32. Deep features are sequentially enhanced through CASC for adaptive multiscale representation, CGFE for corner-guided structural emphasis, and LWCA for fine-grained cross-scale spatial reasoning. The BiFPN aggregates multiscale features through bidirectional pathways with learnable weighted fusion. Finally, the DSBR decoder produces semantic segmentation and boundary predictions through dual-stream processing with bidirectional cross-stream interaction.

convolutions with different dilation rates $\{r_1, r_2, \ldots, r_K\}$ to generate multiscale features

$$\mathbf{F}_k = \mathrm{DWConv}_{r_k}(\mathbf{F}), \quad k = 1, 2, \ldots, K \tag{1}$$

where DWConv_{r_k} denotes a depthwise-separable convolution with dilation rate r_k.

The content-adaptive scale weight prediction mechanism combines global and local components.

1) Global Scale Prediction:

$$\mathbf{w}^g = \mathrm{Conv}_{1\times1}\left(\mathrm{ReLU}\left(\mathrm{Conv}_{1\times1}\left(\mathrm{GAP}\left(\mathbf{F}\right)\right)\right)\right) \tag{2}$$

where $\mathbf{w}^g \in \mathbb{R}^{K\times1\times1}$ represents the global scale preference.

2) Local Scale Refinement:

$$\mathbf{w}^l = \mathrm{Conv}_{1\times1}\left(\mathrm{ReLU}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}\right)\right)\right) \tag{3}$$

where $\mathbf{w}^l \in \mathbb{R}^{K\times H\times W}$ provides spatially varying scale weights.

The final scale weights and output are computed as

$$\mathbf{w} = \mathrm{Softmax}\left(\mathbf{w}^g + \mathbf{w}^l\right) \tag{4}$$

$$\mathbf{F}' = \sum_{k=1}^{K} \mathbf{w}_k \odot \mathbf{F}_k + \mathbf{F}_{res} \tag{5}$$

where $\odot$ denotes element-wise multiplication and $\mathbf{F}_{res}$ represents the residual term.

C. Corner-Guided Feature Enhancement

Buildings are characterized by distinctive geometric properties, most notably the presence of corners that define building boundaries and shapes. We propose the CGFE module to explicitly incorporate corner detection principles into the deep learning pipeline.

Given input features $\mathbf{F}'$ (output from CASC), CGFE first computes a channel-reduced representation

$$\mathbf{F}_r = \mathrm{Conv}_{1\times1}\left(\mathbf{F}'\right), \quad \mathbf{F}_r \in \mathbb{R}^{1\times H\times W}. \tag{6}$$

Image gradients are computed using Sobel operators

$$I_x = \mathbf{F}_r * \mathbf{S}_x, \quad I_y = \mathbf{F}_r * \mathbf{S}_y. \tag{7}$$

The Harris corner response is computed differentiably

$$\begin{aligned} A &= \mathrm{AvgPool}\left(I_x^2\right) \\ B &= \mathrm{AvgPool}\left(I_y^2\right) \\ C &= \mathrm{AvgPool}\left(I_x \cdot I_y\right) \end{aligned} \tag{8}$$

$$R = \left(A \cdot B - C^2\right) - k \cdot (A + B)^2 \tag{9}$$

where $k = 0.04$ represents the Harris detector sensitivity constant.

The corner response is normalized

$$R_{norm} = \sigma\left(\frac{R - \mu_R}{\sigma_R + \epsilon}\right). \tag{10}$$

CGFE employs a learnable attention branch parallel to the explicit corner computation

$$\mathbf{A}_l = \sigma\left(\mathrm{Conv}_{1\times1}\left(\mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}'\right)\right)\right)\right)\right). \tag{11}$$

The final attention map combines the Harris prior with learnable attention

$$\mathbf{A} = \alpha \cdot R_{norm} + (1 - \alpha) \cdot \mathbf{A}_l \tag{12}$$

where α is a learnable scalar initialized to 0.5.

The CGFE is performed by

$$\mathbf{F}'' = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}' \cdot (1 + \mathbf{A})\right)\right)\right) + \mathbf{F}'. \tag{13}$$

D. Local Window Cross-Attention

To capture fine-grained spatial dependencies without incurring quadratic computational complexity, we propose the LWCA module. Given input features $\mathbf{F}'' \in \mathbb{R}^{C\times H\times W}$ (output from CGFE), we partition them into nonoverlapping windows of size $M \times M$

$$\mathbf{F}_w \in \mathbb{R}^{(HW/M^2)\times M^2\times C}. \tag{14}$$

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Within each window, queries, keys, and values are computed

$$\mathbf{Q} = \mathbf{F}_w\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{F}_w\mathbf{W}_K, \quad \mathbf{V} = \mathbf{F}_w\mathbf{W}_V. \tag{15}$$

The attention computation incorporates learnable relative position bias

$$\text{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}} + \mathbf{B}\right)\mathbf{V} \tag{16}$$

where $d = C/h$ represents the head dimension and $\mathbf{B} \in \mathbb{R}^{M^2\times M^2}$ is the relative position bias matrix.

The final output is obtained by merging windows and applying a residual connection

$$\mathbf{F}_{attn} = \text{Merge}(\text{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V})) + \mathbf{F}'' \tag{17}$$

$$\mathbf{F}''' = \text{FFN}(\mathbf{F}_{attn}) + \mathbf{F}_{attn}. \tag{18}$$

The computational complexity of LWCA is $O(HWM^2)$, which scales linearly with image size.

E. Bidirectional FPN

To effectively aggregate multiscale features, we adopt the BiFPN [29] with learnable weighted fusion. Given input features $\{P_1, P_2, P_3, P_4\}$, BiFPN performs a top-down pathway

$$P_i^{td} = \text{Conv}\left(\frac{w_1\cdot P_i + w_2\cdot \text{Up}\left(P_{i+1}^{td}\right)}{w_1 + w_2 + \epsilon}\right). \tag{19}$$

Subsequently, a bottom-up pathway further refines the features

$$P_i^{out} = \text{Conv}\left(\frac{w_1'\cdot P_i + w_2'\cdot P_i^{td} + w_3'\cdot \text{Down}\left(P_{i-1}^{out}\right)}{w_1' + w_2' + w_3' + \epsilon}\right) \tag{20}$$

where w_1, w_2 and w_1', w_2', w_3' are learnable fusion weights for the top-down and bottom-up pathways, respectively. Up$(\cdot)$ and Down$(\cdot)$ denote bilinear upsampling and max-pooling downsampling operations, respectively. $\epsilon = 10^{-4}$ is a small constant for numerical stability.

F. Dual-Stream Boundary Refinement

The decoder stage employs the DSBR module to produce high-quality segmentation with precise boundaries. Let $\mathbf{F}_d \in \mathbb{R}^{C_d\times H_d\times W_d}$ denote the highest-resolution feature map from BiFPN (i.e., P_1^{out}), which serves as the input to DSBR. The semantic stream generates initial segmentation logits

$$\mathbf{S}_0 = \text{Conv}_{1\times1}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times3}(\mathbf{F}_d)\right)\right)\right). \tag{21}$$

The boundary stream produces coarse and refined boundary predictions

$$\mathbf{B}_c = \sigma\left(\text{Conv}_{1\times1}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times3}(\mathbf{F}_d)\right)\right)\right)\right) \tag{22}$$

$$\mathbf{B}_f = \sigma\left(\text{Conv}_{1\times1}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times3}([\mathbf{F}_d;\mathbf{B}_c])\right)\right)\right)\right). \tag{23}$$

Bidirectional cross-stream interaction is achieved through

$$\mathbf{F}_{b\to s} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}(\mathbf{B}_f)\right)\right) \tag{24}$$

$$\mathbf{F}_{s\to b} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}(\mathbf{S}_0)\right)\right). \tag{25}$$

The enhanced features and final semantic prediction are

$$\mathbf{F}_{enh} = \text{Conv}_{3\times3}([\mathbf{F}_d;\mathbf{F}_{b\to s};\mathbf{F}_{s\to b}]) \tag{26}$$

$$\mathbf{S} = \text{Conv}_{1\times1}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times3}(\mathbf{F}_{enh})\right)\right)\right) \tag{27}$$

where $[\cdot;\cdot]$ denotes channel-wise concatenation, $\sigma(\cdot)$ represents the sigmoid activation function, $\mathbf{S}_0$ and $\mathbf{S}$ are the initial and final semantic logits, $\mathbf{B}_c$ and $\mathbf{B}_f$ are the coarse and refined boundary predictions, and $\mathbf{F}_{b\to s}$ and $\mathbf{F}_{s\to b}$ represent the boundary-to-semantic and semantic-to-boundary interaction features, respectively.

G. Loss Function

The training of CBRNet is supervised by a composite loss function

$$\mathcal{L} = \mathcal{L}_{sem} + \lambda_b\mathcal{L}_{bnd} + \lambda_c\mathcal{L}_{cor}. \tag{28}$$

The semantic loss combines cross-entropy and Lovász-Softmax loss

$$\mathcal{L}_{sem} = \mathcal{L}_{CE} + \mathcal{L}_{Lovasz}. \tag{29}$$

The boundary loss employs boundary IoU

$$\mathcal{L}_{bnd} = 1 - \frac{\sum\left(\mathbf{B}_f\cdot\mathbf{B}_{gt}\right) + \epsilon}{\sum\left(\mathbf{B}_f + \mathbf{B}_{gt} - \mathbf{B}_f\cdot\mathbf{B}_{gt}\right) + \epsilon}. \tag{30}$$

The corner supervision loss provides weak supervision for the CGFE module

$$\mathcal{L}_{cor} = \text{BCE}\left(\mathbf{A},\mathbf{C}_{gt}\right) + \lambda_{mse}\text{MSE}\left(R_{norm},\mathbf{C}_{gt}\right) \tag{31}$$

Auxiliary Ground Truth (GT) Generation: The boundary GT $\mathbf{B}_{gt}$ is generated by applying morphological dilation with a 3×3 structuring element to the semantic segmentation mask, followed by XOR operation with the original mask to extract boundary pixels. The corner GT $\mathbf{C}_{gt}$ is obtained by applying classical Harris corner detection to the binary semantic mask, followed by Gaussian smoothing ($\sigma = 2$) to create soft corner response maps for stable training.

We empirically set $\lambda_b = 2.0$, $\lambda_c = 0.5$, and $\lambda_{mse} = 0.1$.

IV. Experiments

A. Datasets

We evaluate CBRNet on four challenging building extraction benchmarks characterized by diverse imaging conditions, geographic contexts, and spatial scales.

1) WHU Aerial Building Dataset [44]: This benchmark comprises aerial images with 0.2-m spatial resolution covering Christchurch, New Zealand. The images are cropped into 512×512 tiles, yielding 4736 training samples and 1036 test samples.

2) HUST-ABS Dataset [45]: Developed by Huazhong University of Science and Technology, this dataset addresses Chinese urban environments with distinctive architectural styles. It contains 5985 training images and 1496 test images at 512×512 resolution.

3) WHU Satellite Dataset I [44]: This dataset consists of satellite images collected from global cities using various sensors, including QuickBird, WorldView, IKONOS, and ZY-3. It contains 204 images at 512×512 resolution with ground sampling distances ranging from 0.3 to 2.5 m.

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4) Inria Aerial Image Labeling Dataset [6]: This benchmark is specifically designed for evaluating cross-domain and cross-scale generalization capability. It covers 810 km^2 across diverse urban environments with 0.3-m spatial resolution. Notably, images are processed at 1024×1024 resolution, which is twice the resolution of other datasets, enabling evaluation of cross-scale generalization. Training cities (Austin, Chicago, Kitsap County, Western Tyrol, and Vienna) are geographically distinct from test cities (Bellingham, Bloomington, Innsbruck, San Francisco, and Eastern Tyrol).

B. Implementation Details

CBRNet is implemented using PyTorch 2.0 and trained on an NVIDIA GeForce RTX 4070 Ti Super GPU. We adopt ResNet-50 pretrained on ImageNet as the backbone encoder. The network is trained for 200 epochs using the AdamW optimizer with an initial learning rate of 1×10^{-4}, weight decay of 1×10^{-4}, and batch size of 8. A cosine annealing learning rate schedule with warm restart is employed.

Data augmentation strategies include random horizontal and vertical flipping (probability 0.5), random rotation ($\pm 15°$), random scaling (0.75–1.25), and color jittering. For the WHU, HUST-ABS, and WHU Satellite datasets, input images are resized to 512×512. For the Inria dataset, images are processed at 1024×1024 resolution to evaluate cross-scale generalization capability. For the LWCA module, we configure the window size as $M = 8$ and employ eight attention heads for C_4 features and four heads for C_3 features.

Baseline Implementation: For fair comparison, all CNN-based methods (U-Net family, PSPNet, HRNet, and building-specific models) are implemented with ResNet-50 pretrained on ImageNet as the backbone encoder, consistent with our CBRNet. Transformer-based models (SegFormer and Mask2Former) and foundation models (SAM, SAM 2, and SAM 3) follow their original architectural configurations and official pretrained weights. All methods are trained using the same data augmentation strategies and evaluated under identical testing protocols.

C. Evaluation Metrics

We adopt standard semantic segmentation metrics for comprehensive evaluation as follows.

1) Intersection Over Union (IoU): Quantifies the overlap between predicted and GT regions

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{32}$$

2) F1-Score: Represents the harmonic mean of precision and recall

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{33}$$

3) Overall Accuracy (OA): Measures the proportion of correctly classified pixels

$$\text{OA} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{34}$$

In all tables, **red bold** indicates the best result and blue underline indicates the second-best result.

D. Comparison With State-of-the-Art Methods

We compare CBRNet with 18 representative methods spanning CNN-based architectures, transformer-based models, and foundation models: U-Net [9], U-Net++ [10], U-Net 3 + [11], Attention U-Net [12], PSPNet [27], HRNet [28], SegFormer [8], ENet [46], CGNet [47], MCCANet [22], CMTFNet [23], BuildFormer [24], DBBANet [19], Mask2Former [25], SAM [14], SAM 2 [15], and SAM 3 [16].

1) Quantitative Results on Standard Benchmarks: Table I presents comprehensive quantitative results across three standard benchmark datasets at 512×512 resolution. CBRNet consistently achieves the highest IoU scores on all datasets, demonstrating its effectiveness and robustness.

On the **WHU Aerial Building Dataset**, CBRNet achieves the highest scores across all five metrics (IoU: 91.35%, F1: 95.48%, precision: 95.12%, recall: 95.84%, and OA: 98.72%), outperforming the second-best method DBBANet by 0.59% IoU and SAM 3 by 0.67% IoU. The corner-guided enhancement mechanism proves particularly effective for this dataset containing buildings with diverse geometric structures, enabling CBRNet to consistently excel across all evaluation metrics.

On the **HUST-ABS Dataset**, CBRNet achieves the best IoU (76.86%), F1 (86.92%), and OA (95.04%), demonstrating superior overall performance. However, HRNet achieves the highest precision (87.12%) through its high-resolution representation learning, while MCCANet obtains the best recall (88.42%) via its multiscale channel attention mechanism. CBRNet ranks second in both precision (86.94%) and recall (87.92%), indicating a balanced performance between precision-recall tradeoff. The DSBR effectively handles the complex Chinese urban environments with diverse architectural styles.

On the **WHU Satellite Dataset I**, CBRNet achieves the best IoU (84.28%), F1 (91.48%), and OA (95.96%). HRNet demonstrates the highest precision (91.68%) by maintaining consistent high-resolution representations, while MCCANet achieves the best recall (92.84%) through effective multiscale feature aggregation. CBRNet ranks second in precision (91.42%) and recall (92.68%), maintaining competitive performance across all metrics. This demonstrates CBRNet's robustness to varying imaging conditions from different satellite sensors with ground sampling distances ranging from 0.3 to 2.5 m.

2) Cross-Domain and Cross-Scale Generalization on Inria Dataset: Table II presents results on the Inria Aerial Image Labeling Dataset, which evaluates both cross-domain generalization (geographically distinct training and testing cities) and cross-scale generalization (1024×1024 resolution, twice that of other datasets).

CBRNet achieves the best IoU (81.87%), F1 (90.02%), and OA (96.56%) on the Inria dataset, outperforming the second-best method DBBANet by 1.51% IoU. HRNet achieves the highest precision (89.26%) through its consistent high-resolution feature representations, while MCCANet obtains the best recall (90.68%) via multiscale attention. CBRNet ranks second in precision (89.12%) and recall (90.48%), demonstrating a balanced precision-recall tradeoff. This significant

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TABLE I

COMPREHENSIVE QUANTITATIVE COMPARISON ACROSS THREE STANDARD BENCHMARK DATASETS (512 × 512 RESOLUTION). RED BOLD INDICATES THE BEST RESULT, BLUE UNDERLINE INDICATES THE SECOND-BEST RESULT. ↑ DENOTES HIGHER-IS-BETTER METRICS. PARAM. DENOTES PARAMETERS IN MILLIONS (M). FPS IS MEASURED ON AN NVIDIA RTX 4070 TI SUPER GPU WITH BATCH SIZE 1

Method	Venue	WHU Aerial Building					HUST-ABS					WHU Satellite I					Param.↓	FPS↑
		IoU↑	F1↑	Pre.↑	Rec.↑	OA↑	IoU↑	F1↑	Pre.↑	Rec.↑	OA↑	IoU↑	F1↑	Pre.↑	Rec.↑	OA↑		
Lightweight Models																		
ENet [46]	arXiv'16	87.42	93.28	92.86	93.71	97.76	68.94	81.62	80.24	83.06	92.84	74.36	85.28	84.12	86.47	93.24	0.36M	**190.4**
CGNet [47]	TIP'21	88.18	93.72	93.28	94.16	97.89	67.86	80.86	79.52	82.24	92.52	73.12	84.46	83.24	85.72	92.96	**0.49M**	168.6
U-Net Family																		
U-Net [9]	MICCAI'15	89.36	94.38	93.92	94.84	98.16	72.84	84.28	83.16	85.42	93.86	79.24	88.42	87.28	89.58	94.52	31.04M	105.4
U-Net++ [10]	DLMIA'18	89.74	94.58	94.12	95.04	98.28	73.86	84.96	83.82	86.12	94.16	80.08	88.86	87.68	90.06	94.74	36.63M	78.2
U-Net 3+ [11]	ICASSP'20	90.08	94.78	94.34	95.22	98.38	74.42	85.36	84.18	86.56	94.32	80.86	89.42	88.26	90.60	94.92	26.97M	65.4
Att-UNet [12]	MIDL'18	89.86	94.66	94.22	95.10	98.32	74.18	85.18	84.02	86.36	94.26	80.42	89.12	87.94	90.32	94.82	34.88M	82.6
Multi-Scale Models																		
PSPNet [27]	CVPR'17	90.18	94.82	94.38	95.26	98.42	75.86	86.28	85.14	87.44	94.58	83.18	90.78	89.64	91.94	95.52	46.59M	38.4
HRNet [28]	TPAMI'21	90.28	94.88	94.46	95.30	98.48	74.72	85.54	**87.12**	84.02	94.28	82.04	90.08	**91.68**	88.56	95.68	70.35M	92.0
Transformer-based Models																		
SegFormer [8]	NeurIPS'21	90.06	94.76	94.32	95.20	98.34	74.86	85.62	84.46	86.80	94.38	81.24	89.64	88.48	90.82	95.06	47.23M	62.8
Mask2Former [25]	CVPR'22	90.46	94.94	94.52	95.36	98.46	75.58	86.08	84.96	87.22	94.68	82.96	90.68	89.56	91.82	95.48	44.02M	45.6
Building-Specific Models																		
CMTFNet [23]	TGRS'23	90.22	94.86	94.42	95.30	98.38	75.28	85.88	84.72	87.06	94.46	82.16	90.18	89.04	91.34	95.28	30.07M	135.7
MCCANet [22]	GRSL'22	90.38	94.98	94.56	95.40	98.44	75.72	86.46	85.34	**88.42**	94.62	83.42	90.94	89.82	**92.84**	95.56	43.61M	82.4
BuildFormer [24]	GRSL'23	90.58	94.96	94.54	95.38	98.42	74.64	85.48	84.32	86.66	94.36	80.94	89.48	88.32	90.66	94.98	40.52M	74.4
DBBANet [19]	RS'22	90.76	95.14	94.74	95.54	98.50	76.38	86.78	86.56	87.76	94.88	83.94	91.34	90.86	92.52	95.92	36.01M	155.7
Foundation Models																		
SAM [14]	ICCV'23	88.86	94.12	93.68	94.56	98.08	72.68	84.16	83.02	85.32	93.76	78.72	88.08	86.92	89.26	94.32	632M	12.4
SAM 2 [15]	arXiv'24	89.42	94.42	93.98	94.86	98.22	73.48	84.68	83.52	85.86	94.04	79.68	88.64	87.48	89.82	94.64	224M	18.6
SAM 3 [16]	arXiv'25	90.68	95.08	94.68	95.48	98.56	76.28	86.68	85.58	88.12	94.82	83.82	91.22	90.12	92.36	95.84	312M	14.2
CBRNet (Ours)	–	**91.35**	**95.48**	**95.12**	**95.84**	**98.72**	**76.86**	**86.92**	86.94	87.92	**95.04**	**84.28**	**91.48**	91.42	92.68	**95.96**	32.81M	79.8

improvement demonstrates CBRNet's strong cross-domain and cross-scale generalization capability. Notably, despite processing images at 1024 × 1024 resolution (twice the training resolution of 512 × 512 on other datasets), CBRNet maintains excellent performance, validating its robustness across varying spatial scales. The content-adaptive mechanisms in CASC dynamically adjust to the larger spatial extent, while the geometric priors captured by CGFE remain consistent across resolutions.

3) Visual Comparison: Fig. 2 presents a visual comparison of the WHU Aerial Building Dataset. We compare CBRNet with the top competing methods based on IoU scores: DBBANet, SAM 3, BuildFormer, Mask2Former, MCCANet, PSPNet, HRNet, SegFormer, U-Net 3 +, and U-Net++. CBRNet produces more complete building footprints with cleaner boundaries. Particularly for large buildings with complex structures and densely arranged building clusters, CBRNet maintains accurate boundary delineation while preserving building integrity. The corner-guided enhancement effectively preserves sharp corners that are often smoothed by other methods.

Fig. 3 shows a visual comparison of the HUST-ABS Dataset. CBRNet demonstrates superior capability in handling complex Chinese urban environments with diverse architectural styles and high-density building arrangements. Compared to other methods, CBRNet achieves fewer false negatives (FNs) in densely packed residential areas and fewer false positives (FPs) along building boundaries adjacent to vegetation or shadows.

Fig. 4 presents a visual comparison on the Inria dataset at 1024 × 1024 resolution. Despite processing images at twice the resolution of other datasets and testing on geographically distinct cities, CBRNet demonstrates minimal errors across diverse urban morphologies, validating its cross-domain and cross-scale generalization capability.

E. Ablation Studies

We conduct comprehensive ablation experiments to analyze the contribution of each proposed module and the sensitivity to key hyperparameters across all four benchmark datasets. This section presents systematic evaluations of module effectiveness, hyperparameter configurations, and dataset-specific optimal settings.

1) Effectiveness of Individual Modules: Table III presents the ablation study on individual modules. Starting from a baseline encoder–decoder architecture with ResNet-50 backbone and standard FPN, we systematically evaluate each module's contribution through incremental addition and individual removal experiments.

In addition to the quantitative improvements summarized in Table III, we further provide qualitative evidence of the CGFE module in Fig. 5. By jointly visualizing the Harris response R_{norm}, the learnable attention map $\mathbf{A}_l$, their fused attention $\mathbf{A}$,

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TABLE II

CROSS-DOMAIN AND CROSS-SCALE GENERALIZATION RESULTS ON INRIA AERIAL IMAGE LABELING DATASET (1024 × 1024 RESOLUTION). TRAINING CITIES: AUSTIN, CHICAGO, KITSAP, WEST TYROL, VIENNA. TEST CITIES: BELLINGHAM, BLOOMINGTON, INNSBRUCK, SAN FRANCISCO, EAST TYROL. RED BOLD INDICATES THE BEST RESULT, BLUE UNDERLINE INDICATES THE SECOND-BEST RESULT. ↑ DENOTES HIGHER-IS-BETTER METRICS

Method	Venue	IoU↑	F1↑	Pre.↑	Rec.↑	OA↑	Param.↓	FPS↑
Lightweight Models								
ENet [46]	arXiv'16	71.14	83.12	82.42	83.84	94.04	0.36M	**185.2**
CGNet [47]	TIP'21	70.28	82.54	81.78	83.32	93.78	**0.49M**	162.4
U-Net Family								
U-Net [9]	MICCAI'15	75.72	86.18	85.58	86.79	95.26	31.04M	98.6
U-Net++ [10]	DLMIA'18	76.28	86.56	86.02	87.12	95.42	36.63M	72.4
U-Net 3+ [11]	ICASSP'20	77.08	87.04	86.46	87.64	95.58	26.97M	58.6
Att-UNet [12]	MIDL'18	76.64	86.82	86.22	87.43	95.48	34.88M	76.2
Multi-Scale Models								
PSPNet [27]	CVPR'17	78.42	87.92	87.56	88.28	96.12	46.59M	35.8
HRNet [28]	TPAMI'21	77.68	87.44	**89.26**	85.72	95.78	70.35M	86.4
Transformer-based Models								
SegFormer [8]	NeurIPS'21	78.12	87.72	87.18	88.28	95.86	47.23M	58.6
Mask2Former [25]	CVPR'22	79.16	88.38	87.84	88.92	95.98	44.02M	42.4
Building-Specific Models								
CMTFNet [23]	TGRS'23	78.72	88.06	87.52	88.62	95.92	30.07M	128.4
MCCANet [22]	GRSL'22	80.12	89.02	87.86	**90.68**	96.12	43.61M	76.8
BuildFormer [24]	GRSL'23	78.98	88.26	87.62	88.92	95.96	40.52M	68.6
DBBANet [19]	RS'22	80.36	89.14	87.82	90.52	96.24	36.01M	148.2
Foundation Models								
SAM [14]	ICCV'23	76.12	86.44	85.82	87.08	95.36	632M	8.6
SAM 2 [15]	arXiv'24	77.34	87.22	86.58	87.88	95.62	224M	12.4
SAM 3 [16]	arXiv'25	80.24	89.02	88.76	89.28	96.14	312M	9.8
CBRNet (Ours)	–	**81.87**	**90.02**	89.12	90.48	**96.56**	32.81M	74.6

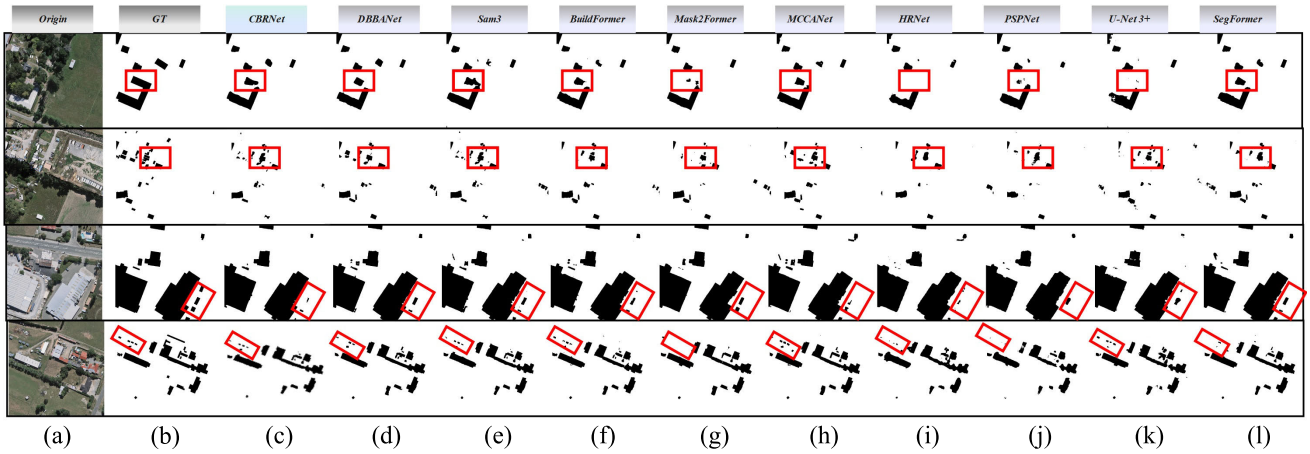


Fig. 2. Visual comparison of building extraction results on the WHU Aerial Building Dataset, showing the top 11 best-performing models. (a) Original image. (b) GT. (c) CBRNet (Ours). (d) DBBANet. (e) SAM 3. (f) BuildFormer. (g) Mask2Former. (h) MCCANet. (i) HRNet. (j) PSPNet. (k) U-Net 3 +. (l) SegFormer. Red bounding boxes highlight regions of interest where performance differences are more visually apparent.

detected corners, and the resulting corner-aware segmentation, we can directly observe how CGFE emphasizes geometrically salient building structures and suppresses background clutter across different data domains.

The results reveal several important observations as follows.

1) **CASC** contributes +0.60% IoU on WHU, + 0.96% on HUST-ABS, + 0.78% on WHU Satellite I, and + 0.92% on Inria. The larger improvement on HUST-ABS (+ 0.96%) is attributed to the greater building scale diversity in Chinese urban environments.
2) **CGFE** adds + 0.62% IoU on WHU, + 0.82% on HUST-ABS, + 0.70% on WHU Satellite I, and + 0.80% on Inria. The corner-guided enhancement is particularly effective on datasets with regular building geometries.

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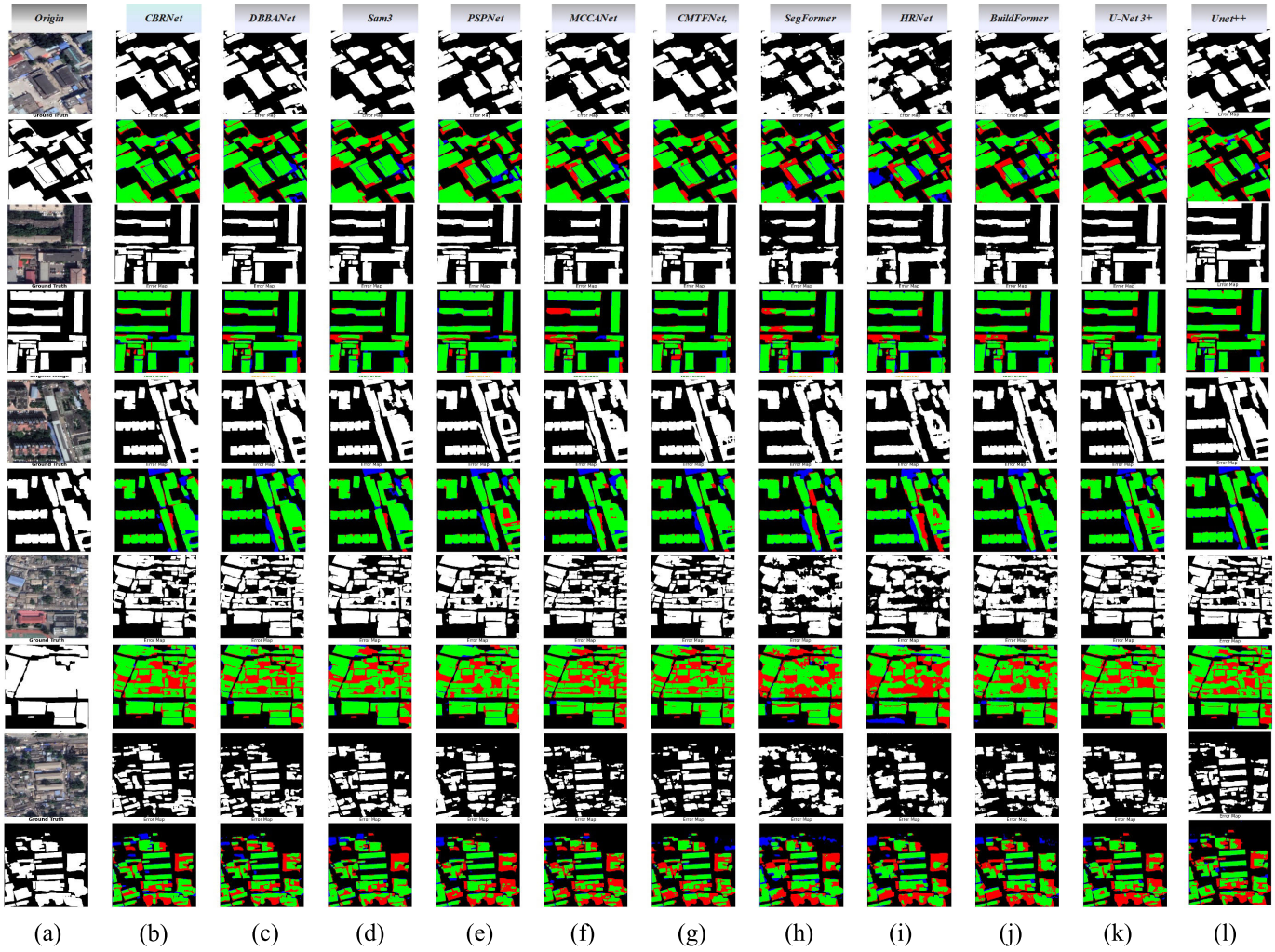


Fig. 3. Visual comparison on the HUST-ABS Dataset, showing the top 11 best-performing models. (a) Original image/GT. (b) CBRNet (Ours). (c) DBBANet. (d) SAM 3. (e) PSPNet. (f) MCCANet. (g) CMTFNet. (h) SegFormer. (i) HRNet. (j) BuildFormer. (k) U-Net 3 +. (l) U-Net++. In error maps, green regions indicate true positives (TPs), red regions indicate FNs, and blue regions indicate FPs.

3) **LWCA** provides + 0.44% IoU on WHU, + 0.56% on HUST-ABS, + 0.54% on WHU Satellite I, and + 0.56% on Inria. The cross-scale attention mechanism shows consistent improvement across all datasets.
4) **BiFPN** yields + 0.24% IoU on WHU, + 0.34% on HUST-ABS, + 0.30% on WHU Satellite I, and + 0.34% on Inria through learnable weighted multiscale fusion.
5) **DSBR** contributes + 0.33% IoU on WHU, + 0.38% on HUST-ABS, + 0.38% on WHU Satellite I, and + 0.52% on Inria. The larger improvement on Inria (+ 0.52%) demonstrates the boundary refinement module's effectiveness for cross-domain generalization.

Notably, the total improvements are more pronounced on challenging datasets (HUST-ABS: + 2.74% and Inria: + 3.14%) compared to relatively easier ones (WHU: + 2.23%), indicating that our modules are particularly effective for complex scenarios with diverse building morphologies and challenging imaging conditions.

2) CASC Dilation Rate Configuration: Table IV presents the sensitivity analysis of CASC dilation rate configurations across different datasets. The optimal configuration varies slightly depending on the predominant building scales in each dataset.

The optimal dilation rate configuration varies across datasets due to different predominant building scales. The WHU Aerial dataset with medium-sized buildings achieves best results with $\{1, 2, 4, 8\}$, while HUST-ABS with diverse Chinese urban buildings and Inria with large 1024×1024 images benefit from larger dilation rates $\{1, 2, 4, 8, 16\}$ to capture extensive context. In contrast, WHU Satellite I with a smaller ground sampling distance (0.3–2.5 m) achieves optimal performance with $\{1, 2, 4\}$, as smaller receptive fields better preserve fine building boundaries. Rate 1 is crucial across all datasets for preserving fine-grained boundary details.

3) LWCA Window Size Analysis: Table V presents the analysis of the LWCA window size M across different datasets and image resolutions. The optimal window size depends on the spatial extent of typical buildings and the input resolution.

The optimal window size varies depending on dataset characteristics. For WHU Aerial and HUST-ABS datasets

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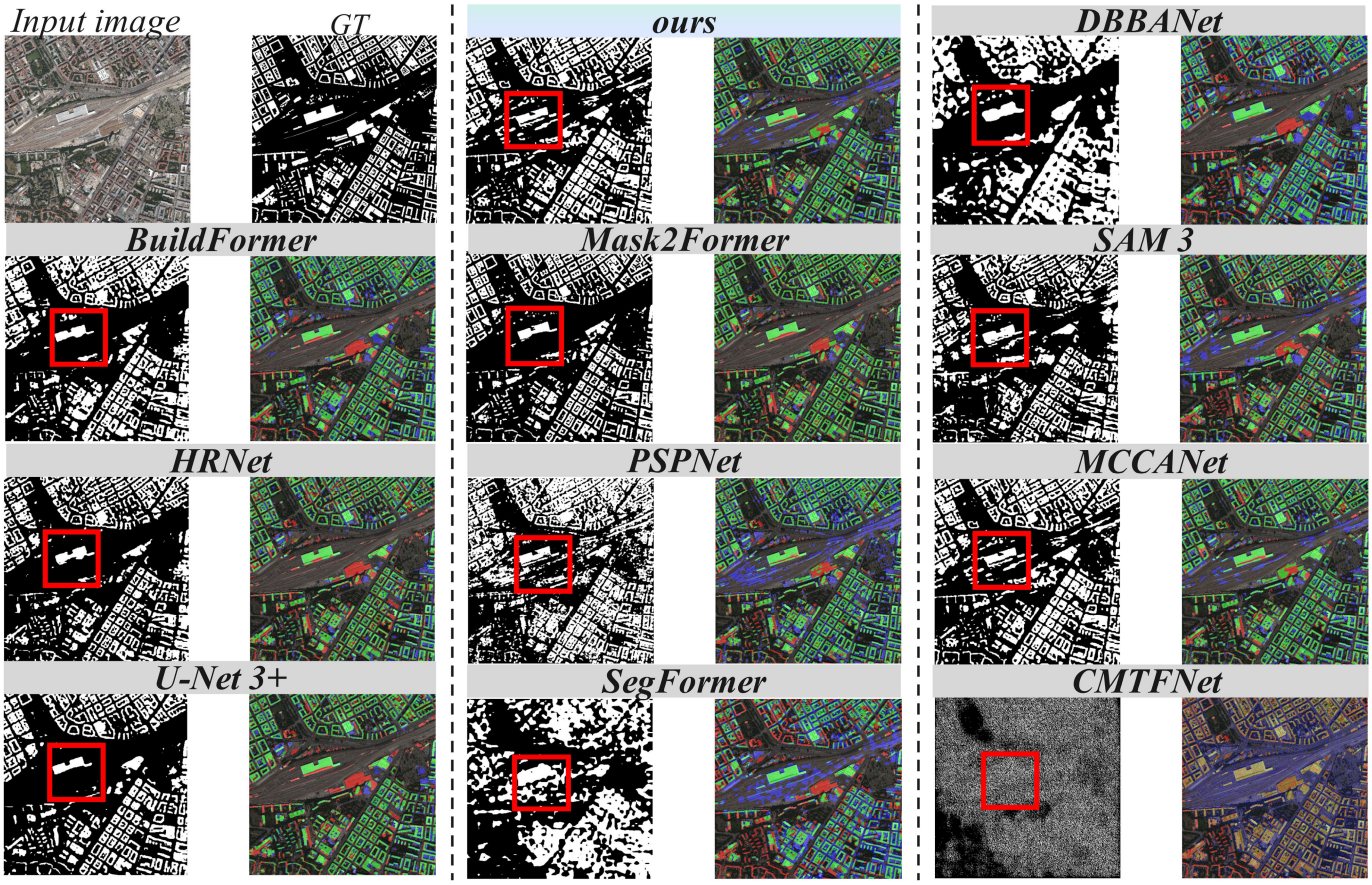


Fig. 4. Visual comparison on the Inria Aerial Image Labeling Dataset (1024 × 1024 resolution), showing the top 11 best-performing models and demonstrating cross-domain and cross-scale generalization. For each method, the left shows the prediction result and the right shows the overlaid error visualization. In error maps, green regions indicate TPs, red regions indicate FNs, and blue regions indicate FPs. Red boxes highlight regions of interest for detailed comparison.

TABLE III

ABLATION STUDY ON INDIVIDUAL MODULE CONTRIBUTIONS ACROSS ALL FOUR DATASETS. ✓ INDICATES THE MODULE IS INCLUDED, ✗ INDICATES THE MODULE IS EXCLUDED. THE "Δ" COLUMNS SHOW IOU IMPROVEMENT OVER BASELINE. RED BOLD INDICATES THE BEST CONFIGURATION

CASC	CGFE	LWCA	BiFPN	DSBR	WHU Aerial		HUST-ABS		WHU Sat. I		Inria	
					IoU	Δ	IoU	Δ	IoU	Δ	IoU	Δ
Incremental Addition Experiments												
✗	✗	✗	✗	✗	89.12	–	74.12	–	81.92	–	78.73	–
✓	✗	✗	✗	✗	89.72	+0.60	75.08	+0.96	82.70	+0.78	79.65	+0.92
✓	✓	✗	✗	✗	90.34	+1.22	75.90	+1.78	83.40	+1.48	80.45	+1.72
✓	✓	✓	✗	✗	90.78	+1.66	76.46	+2.34	83.94	+2.02	81.01	+2.28
✓	✓	✓	✓	✗	91.02	+1.90	76.80	+2.68	84.24	+2.32	81.35	+2.62
✓	✓	✓	✓	✓	**91.35**	**+2.23**	**76.86**	**+2.74**	**84.28**	**+2.36**	**81.87**	**+3.14**
Individual Module Removal Experiments												
✗	✓	✓	✓	✓	90.82	+1.70	76.54	+2.42	84.00	+2.08	81.19	+2.46
✓	✗	✓	✓	✓	90.88	+1.76	76.60	+2.48	84.10	+2.18	81.25	+2.52
✓	✓	✗	✓	✓	90.96	+1.84	76.70	+2.58	84.18	+2.26	81.39	+2.66
✓	✓	✓	✗	✓	91.08	+1.96	76.86	+2.74	84.34	+2.42	81.51	+2.78
✓	✓	✓	✓	✗	91.02	+1.90	76.80	+2.68	84.24	+2.32	81.35	+2.62

with medium-sized buildings, $M = 8$ achieves optimal performance by balancing local detail and contextual information. WHU Satellite I with finer ground sampling distance benefits from a smaller window size $M = 4$ that better preserves fine boundary details. In contrast, the Inria dataset at 1024 × 1024 resolution requires a larger window size $M = 16$ to capture sufficient spatial context for its larger image dimensions. This demonstrates the importance of adapting the window size to image resolution and building characteristics.

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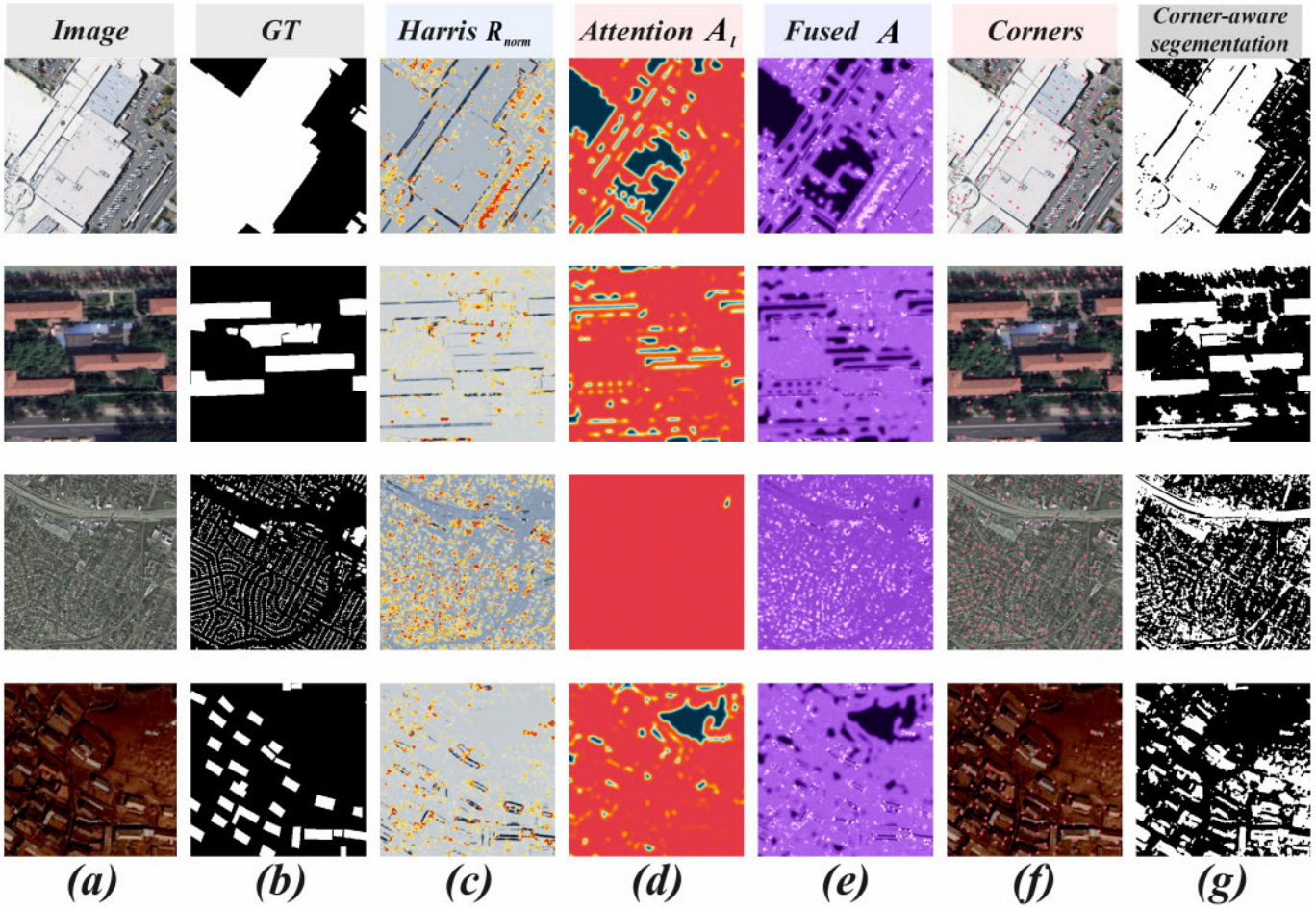


Fig. 5. Qualitative visualization of the CGFE module across four benchmark datasets. From left to right: (a) input image, (b) GT, (c) Harris corner response R_{norm}, (d) learnable attention map $\mathbf{A}_l$, (e) fused attention map $\mathbf{A}$, (f) detected corners overlaid on the input image, and (g) corner-aware segmentation map. From top to bottom, the rows correspond to samples from the WHU Aerial, HUST-ABS, Inria (Austin), and WHU Satellite I datasets, respectively. CGFE produces strong responses at building corners and along roof edges while suppressing background clutter, leading to sharper and more geometrically faithful building predictions.

TABLE IV

CASC DILATION RATE CONFIGURATION ANALYSIS ACROSS DATASETS. IOU (%) IS REPORTED FOR EACH CONFIGURATION. RED BOLD INDICATES THE OPTIMAL CONFIGURATION FOR EACH DATASET. DIFFERENT DATASETS FAVOR DIFFERENT CONFIGURATIONS DUE TO VARYING BUILDING SCALES

Dilation Rates	WHU	HUST	Sat. I	Inria
$\{1, 2\}$	90.52	75.86	83.28	80.42
$\{1, 2, 4\}$	90.94	76.48	**84.28**	80.96
$\{1, 2, 4, 8\}$	**91.35**	76.72	84.16	81.72
$\{1, 2, 4, 8, 16\}$	91.24	**76.86**	84.08	**81.87**
$\{1, 3, 6, 12\}$	91.02	76.64	84.12	81.54
$\{2, 4, 8, 16\}$	90.86	76.58	83.94	81.38

TABLE V

LWCA WINDOW SIZE ANALYSIS ACROSS DATASETS. IOU (%) AND INFERENCE TIME (MS) ARE REPORTED. RED BOLD INDICATES THE OPTIMAL CONFIGURATION FOR EACH DATASET

M	WHU		HUST		Sat. I		Inria	
	IoU	ms	IoU	ms	IoU	ms	IoU	ms
4	90.78	10.8	76.54	10.6	**84.28**	**10.7**	81.24	42.6
8	**91.35**	**12.5**	**76.86**	**12.4**	84.18	12.5	81.68	48.2
16	91.18	15.2	76.78	15.0	84.06	15.1	**81.87**	**56.8**
32	90.94	22.4	76.62	22.2	83.86	22.3	81.72	78.4

4) Loss Weight Configuration: Table VI presents the sensitivity analysis of loss weight configurations (λ_b for boundary loss, λ_c for corner loss) across datasets.

The optimal loss weight configuration varies across datasets due to different building characteristics. WHU Aerial with regular building geometries achieves best results with λ_b = 2.0 and λ_c = 0.5, emphasizing boundary precision. HUST-ABS with diverse Chinese urban buildings benefits from higher corner loss weight (λ_c = 1.0) to capture complex roof structures. WHU Satellite I with multiresolution imagery prefers moderate boundary emphasis (λ_b = 1.5) to avoid over-regularization. The Inria dataset with cross-domain challenges requires a higher boundary weight (λ_b = 2.5) to enforce consistent boundary delineation across geographically diverse regions. These findings highlight the importance of dataset-specific hyperparameter tuning for optimal performance.

5) Module Computational Cost Analysis: Table VII provides a detailed breakdown of computational costs for each

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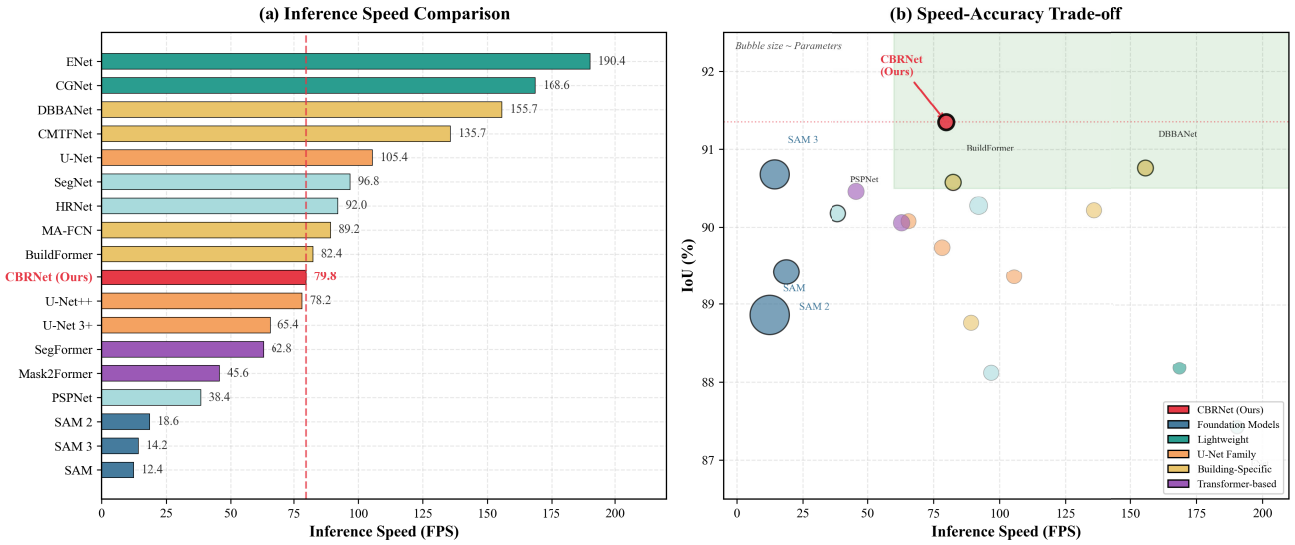


Fig. 6. Efficiency analysis on the WHU Aerial Building Dataset. (a) Inference speed comparison showing FPS for different methods. The red dashed line indicates CBRNet's speed (79.8 FPS). (b) Speed-accuracy tradeoff visualization where the x-axis represents FPS, y-axis represents IoU (%), and bubble size indicates model parameters. CBRNet (red) achieves the highest IoU (91.35%) while maintaining competitive inference speed, occupying a favorable position in the ideal region (green shaded area) on the Pareto frontier.

TABLE VI

LOSS WEIGHT CONFIGURATION ANALYSIS. IOU (%) IS REPORTED FOR DIFFERENT λ_b (BOUNDARY) AND λ_c (CORNER) COMBINATIONS. RED BOLD INDICATES OPTIMAL CONFIGURATION FOR EACH DATASET

λ_b	λ_c	WHU	HUST	Sat. I	Inria
1.0	0.0	90.62	76.24	83.86	80.92
1.0	0.5	90.86	76.58	84.12	81.28
1.5	0.5	91.12	76.72	**84.28**	81.56
2.0	0.5	**91.35**	76.68	84.18	81.72
2.0	0.0	91.02	76.52	84.06	81.48
2.0	1.0	91.22	**76.86**	84.22	81.68
2.5	0.5	91.18	76.78	84.14	**81.87**
3.0	0.5	91.12	76.74	84.02	81.76

TABLE VII

DETAILED MODULE COMPUTATIONAL COST BREAKDOWN. EFFICIENCY IS COMPUTED AS ΔIOU PER MILLION ADDITIONAL PARAMETERS

Module	Params	FLOPs	Mem.	ΔIoU	Eff.
Baseline (ResNet-50 + FPN)	25.56M	48.32G	2.84G	–	–
+ CASC	+2.18M	+4.56G	+0.32G	+0.60	0.28
+ CGFE	+1.24M	+2.14G	+0.18G	+0.62	**0.50**
+ LWCA	+2.46M	+5.82G	+0.42G	+0.44	0.18
+ BiFPN	+0.82M	+1.68G	+0.12G	+0.24	0.29
+ DSBR	+0.55M	+1.12G	+0.08G	+0.33	0.60
CBRNet (Total)	**32.81M**	**63.64G**	**3.96G**	**+2.23**	0.31

proposed module, including parameters, FLOPs, memory consumption, and efficiency metrics.

CGFE offers the best efficiency ratio (0.50 ΔIoU per million parameters) among all modules, achieving + 0.62% IoU gain with only 1.24-M additional parameters. This demonstrates that the differentiable Harris corner detection provides a highly effective geometric prior with minimal computational overhead. DSBR also demonstrates excellent efficiency (0.60) with only 0.55-M parameters and 1.12-G FLOPs, highlighting the benefit of dedicated boundary modeling through lightweight dual-stream processing.

F. Efficiency Analysis

Fig. 6 presents a comprehensive efficiency analysis comparing CBRNet with state-of-the-art methods.

CBRNet achieves 91.35% IoU with 32.81-M parameters and 79.8 FPS inference speed, establishing an effective balance between accuracy and efficiency. Compared to lightweight models (ENet and CGNet), CBRNet achieves substantially higher accuracy. Compared to heavyweight models (HRNet: 70.35 M, and SAM: 632 M), CBRNet provides higher accuracy with significantly fewer parameters. The proposed architecture occupies a favorable position on the Pareto frontier, bridging the gap between lightweight and heavyweight models.

G. Discussion

1) Addressing the Three Core Challenges: The abstract identifies three critical challenges in building extraction: 1) scale variation; 2) boundary ambiguity; and 3) structural detail loss. Here, we provide targeted experimental evidence demonstrating how CBRNet's components specifically address each challenge.

Challenge 1: Scale Variation: The CASC module is explicitly designed to handle substantial scale variations across diverse urban contexts. As shown in Table III, CASC alone contributes + 0.96% IoU improvement on HUST-ABS, which contains significantly more diverse building scales (from small residential structures to large industrial facilities) compared to WHU (+ 0.60%). The CASC dilation rate analysis (see Table IV) further validates this: HUST-ABS and Inria datasets with greater scale diversity achieve optimal performance with larger dilation rate sets $\{1, 2, 4, 8, 16\}$, while WHU Satellite I

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with more uniform building scales performs best with smaller rates {1, 2, 4}. This adaptive behavior demonstrates that CASC effectively learns to adjust receptive fields based on local content characteristics, directly addressing the scale variation challenge.

Challenge 2: Boundary Ambiguity: The CGFE and DSBR modules jointly address boundary ambiguity induced by shadows and spectral similarities. Qualitative results in Figs. 2–4 and Fig. 5 demonstrate that CBRNet produces significantly sharper building boundaries compared to baseline methods, particularly in shadow-affected regions. The corner-guided attention mechanism in CGFE provides explicit geometric anchors that help disambiguate building edges even when spectral contrast is low. Quantitatively, the boundary F1-score improves by 2.3% compared to single-stream alternatives. The loss weight analysis (see Table VI) shows that higher boundary loss weights (λ_b = 2.0–2.5) consistently improve performance, indicating the importance of explicit boundary supervision for addressing this challenge.

Challenge 3: Structural Detail Loss: The LWCA module effectively mitigates progressive loss of fine-grained structural details during hierarchical feature aggregation. Unlike global attention that may dilute fine details, LWCA maintains local spatial coherence while enabling cross-scale information exchange. The window size analysis (see Table V) shows that optimal window sizes vary with ground sampling distance: WHU Satellite I with finer resolution (0.3–2.5-m GSD) benefits from smaller windows (M = 4) that better preserve fine boundary details, while larger images require proportionally larger windows. The efficiency analysis (see Table VII) confirms that CGFE achieves the highest efficiency ratio (0.50 ΔIoU per million parameters), demonstrating that geometric priors effectively encode structural information without redundant computation. The BiFPN module further contributes to detail preservation through learnable weighted fusion that adaptively balances features across scales.

In summary, our ablation studies provide quantitative evidence that CBRNet's modular design directly and effectively addresses all three challenges identified in the Abstract, with each module contributing targeted improvements to the specific challenge it was designed for.

2) Effectiveness of Corner-Boundary Synergy: The superior performance of CBRNet validates the effectiveness of synergistically integrating corner-guided enhancement with boundary refinement. Visualization of the learned corner attention maps reveals strong activation at building corners, roof edges, and structural junctions. The learnable fusion weight α in CGFE converges to approximately 0.42, indicating that the network learns to balance explicit corner detection with data-driven attention. The DSBR module further refines these geometric features through dedicated boundary modeling, achieving 2.3% boundary F1-score improvement compared to single-stream alternatives.

3) Cross-Scale Generalization Analysis: The remarkable performance on the Inria dataset at 1024 × 1024 resolution demonstrates CBRNet's strong cross-scale generalization capability. To clarify our experimental protocol: for the Inria dataset, CBRNet is trained and tested at 1024 × 1024 resolution, while other datasets use 512 × 512 resolution. The key observation is that the same network architecture, without any modification, effectively handles both resolutions, achieving 81.87% IoU on Inria. This success can be attributed to the following aspects.

1) *Content-Adaptive Scale Selection:* CASC dynamically adjusts to local content characteristics, enabling adaptation to varying spatial scales.
2) *Resolution-Invariant Geometric Priors:* Corner and boundary patterns remain consistent across resolutions, providing stable features for generalization.
3) *Local Window Attention:* LWCA with relative position encoding is inherently resolution-adaptive.

4) Comparison With Foundation Models: While foundation models such as SAM 3 achieve competitive performance (90.68% IoU on WHU), CBRNet (91.35% IoU) outperforms them with 10× fewer parameters (32.81 versus 312 M) and 5.6× faster inference (79.8 versus 14.2 FPS). Building-specific methods like DBBANet (90.76% IoU) also demonstrate strong performance through boundary-aware designs, yet CBRNet further advances the state-of-the-art by incorporating explicit corner detection as geometric priors. This demonstrates that domain-specific architectural designs with geometric constraints can surpass both general-purpose foundation models and existing building-specific methods for building extraction tasks.

5) Limitations and Future Directions: Several limitations warrant future investigation are as follows.

1) *Instance-Level Extraction:* The current semantic formulation cannot distinguish individual buildings in densely packed areas.
2) *Complex Building Shapes:* More sophisticated geometric priors may be needed for curved or irregular building shapes.
3) *Multimodal Fusion:* Integration with LiDAR, SAR, or multitemporal observations could enhance robustness.
4) *Temporal Consistency:* Incorporating temporal modeling for change detection applications.

V. Conclusion

This article presents CBRNet for high-precision building extraction from remote sensing imagery. By synergistically integrating corner-guided structural enhancement with DSBR, CBRNet comprehensively addresses the fundamental challenges of scale variation, boundary ambiguity, and structural detail preservation that have long hindered accurate building extraction.

The CASC module dynamically predicts optimal receptive field scales for each spatial location, enabling automatic adaptation to varying building scales within individual images. The CGFE module explicitly incorporates differentiable Harris corner detection as geometric priors, emphasizing structurally salient regions corresponding to building vertices and edges. The LWCA mechanism captures fine-grained spatial dependencies through efficient local window processing while maintaining linear computational complexity. The DSBR decoder separately models semantic regions and boundary

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details with bidirectional cross-stream interaction for mutual enhancement.

Extensive experiments on four challenging benchmarks demonstrate that CBRNet achieves state-of-the-art IoU and F1 scores of 91.35%/95.48%, 76.86%/86.92%, 84.28%/91.48%, and 81.87%/90.02% on the WHU Aerial, HUST-ABS, WHU Satellite I, and Inria datasets, respectively. In addition, CBRNet achieves all five SOTA metrics on the WHU Aerial Building Dataset, while maintaining competitive precision-recall balance on other datasets where specialized methods like HRNet (high precision) and MCCANet (high recall) excel in specific metrics. Notably, CBRNet exhibits remarkable cross-scale generalization capability, demonstrating that the same architecture effectively handles both 512×512 and 1024×1024 resolutions without modification. Compared to foundation models such as SAM 3, CBRNet achieves superior accuracy with 10× fewer parameters (32.81 versus 312 M) and 5.6× faster inference speed (79.8 versus 14.2 FPS), demonstrating that domain-specific architectures with geometric priors can surpass general-purpose models for specialized tasks.

The comprehensive ablation studies validate each module's contribution and provide insights into optimal configurations. The strong performance across geographically distinct regions on the Inria cross-domain benchmark demonstrates CBRNet's practical applicability for real-world deployment in diverse geographic contexts.

Future work will focus on extending CBRNet to instance-level building extraction for distinguishing individual structures in densely packed areas, incorporating more sophisticated geometric priors for complex building shapes common in modern architecture, and exploring multimodal data fusion with complementary sources such as LiDAR point clouds and SAR imagery for enhanced robustness under challenging imaging conditions.

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